Multivariate Decomposition of a Difference in Hazard Rates

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Abstract

We discuss methods for decomposing group differences in empirical rates using model-based approaches that extend traditional Oaxaca-Blinder decomposition methods to generalized linear models. These approaches are useful in the analysis of health disparities because they provide linkages between observed group differences in risk and protective factors and the behavioral responses, or returns to risk, of those factors that can be estimated from statistical models. They provide answers to questions such as how much a disparity would be expected to grow or shrink if the characteristics of one group were substituted with those of another. We aim to replicate a typical analysis and walk the user through the methodology of the approach as well as the steps to carry out a decomposition of a difference in rates of early family formation using data from the 1979 National Longitudinal Survey of Youth. Our accompanying `R` and `Stata` routines provide convenient implementations of multivariate decomposition for a variety of outcome measures.[1]

Introduction

This document provides a description of the methodology, preliminary analysis, and a reproduction of the results in the companion paper on multivariate decomposition for hazard rate models by Powers and Yun (2009).[2] The approach extends the Oaxaca-Blinder (OB) method to decompose group differences in rates in the spirit of past demographic research (i.e., work by Evelyn Kitagawa, 1955 and Prithwis Das Gupta, 1990).[3] And it extends the standard OB decomposition common in labor economics to nonlinear response models.[4]

Background

General decomposition methods such as Oaxaca-Blinder have long been used in labor economics to study gender and racial disparities in earnings with the goal of quantifying the contributions of human capital characteristics as well as the returns to these components on wage differences.[5] OB-type decomposition methods are now in common use in investigations of health disparities and inequities in many fields.

[1] This document is an extended version of class notes used in the course *Longitudinal Data Analysis* (SOC386L/SDS385.9) at the University of Texas at Austin.

[2] Powers, D.A. and M.-S. Yun (2009). "Multivariate Decomposition for Hazard Rate Models." *Sociological Methodology, 39: 233-263.*

[3] Kitagawa, E.M. 1955. "Components of a Difference between Two Rates." *Journal of the American Statistical Association* 50:1168–94; Das Gupta, P. 1993. *Standardization and Decomposition of Rates: A User's Manual.*Washington, DC: U.S. Government Printing Office.

[4] Oaxaca, R.L. 1973. "Male-Female Wage Differentials in Urban Labor Markets," *International Economic Review*, 14: 693-709. Blinder A.S. 1973. "Wage Discrimination: Reduced Form and Structural Variables," *Journal of Human Resources*, 8: 436-455.

[5] see, e.g., Fortin, N., T. Lemieux, S. Firpo., 2010 *Decomposition Methods in Economics*, *Handbook of Labor Economics V4a*, Elsevier B.V. ISSN 0169-7218, doi10.1016/S0169-7218(11)00407-2

The extension of the OB-approach presented in the companion paper applies to individual-level observational event-history data, where a difference in overall hazards (rates) between two groups is decomposed into components that reflect group differences in the distributions of observed characteristics (E) and group differences in effects of these characteristics (C). The difference in rates in part reflects the fact that groups being compared are *compositionally* different, with different distributions on risk and protective factors (X). These factors are observable and perhaps modifiable by policies targeted to reduce disparities between the two groups under study. The effects (β) that these characteristics have on individual behavior and outcomes are unobserved, but are easily estimated by regression-type models applied to event-history data.[6] Distinguishing components of a difference in rates in this way permits additional insight into processes generating disparities that are not readily available from the results of multivariate models and descriptive statistics alone.

The companion paper discusses a feasible approach to obtain components of a difference in the Black-White premarital birth rate based on a piecewise-constant exponential hazard model. In this case, an observed difference in rates can be decomposed into *overall* E and C components, as well as covariate-specific contributions to E and C referred to as a *detailed* decomposition, in which E_k and C_k ($k = 1, \ldots, K$) represent contributions of the kth risk or protective factor to each of the overall components.

Theory

Consider two groups, A and B with overall rates r_A and r_B, characteristics X_A and X_B, with effects β_A and β_B.[7] The difference between group A's and group B's rate can be written as,

$$\begin{aligned} r_A - r_B &= \overline{F(X_A\beta_A)} - \overline{F(X_B\beta_B)} \\ &= \underbrace{\left[\overline{F(X_A\beta_A)} - \overline{F(X_B\beta_A)}\right]}_{E} + \underbrace{\left[\overline{F(X_B\beta_A)} - \overline{F(X_B\beta_B)}\right]}_{C}. \end{aligned} \tag{1}$$

The E component refers to the part of the rate differential owing to differences in *endowments*, characteristics, or composition, usually called the explained component, characteristics component, or composition component. The C component refers to that part of the differential attributable to differences in *coefficients* or effects, often called the unexplained component or coefficients component. In Eq.(1) we have chosen group A as the reference group. Then E reflects a counterfactual comparison of the difference in rates from group A's perspective, where the expected difference in rates, $\overline{F(X_A\beta_A)} - \overline{F(X_B\beta_A)}$, reflects a comparison of group A's and group B's rates if group A's returns to risk were applied to group B's characteristics.[8] The C component reflects the difference in rates, $\overline{F(X_B\beta_A)} - \overline{F(X_B\beta_B)}$, that would prevail if group B was subject to group A's behavioral responses, returns to risk, or effects of X.[9]

The Indexing Problem: The difference in rates shown in Eq.(1) uses group A as the reference. This parameterization is not unique. A decomposition could just as well be carried out from group B's perspective. That is, $E = \overline{F(X_A\beta_B)} - \overline{F(X_B\beta_B)}$ would reflect the difference in rates if group A faced group B's returns to risk. For the C component, group A's characteristics would be assigned to both groups, $\overline{F(X_A\beta_A)} - \overline{F(X_A\beta_B)}$. Although the overall difference in each of these decompositions would remain the same, the components E and C will differ across the two decompositions.

[6] In the field of labor economics the effects of characteristics are often regarded as evidence of discrimination in the wage determination process (i.e., the wage structure component).

[7] We adopt the shorthand, $X\beta = \mathbf{x}_i^T\boldsymbol{\beta}$, where $\mathbf{x}_i$ is a $K \times 1$ vector of characteristics for individual i, and $\boldsymbol{\beta}$ is a $K \times 1$ vector of point estimates from a piecewise constant exponential hazard model. The "overbar" in Eq.1 indicates the average over the specific sample represented by A and B.

[8] We can think of this as group A's actual rate being compared to group B's counterfactual rate where the returns to risk (β) are those of group A, or as group A's actual rate being compared to group A's counterfactual rate under group B's distribution of risk factors (X).

[9] We can think of $\overline{F(X_B\beta_A)}$ as group B's counterfactual rate under group A's effects. This is compared to group B's actual rate. This conceptualization reflects an average treatment effect on the treated (ATT) in some sense.

Details on the General Approach

In Eq.(1), define

$$F(X_j\beta_j) = \Lambda_{i\ell j} = \Lambda_{0\ell j}e^{X_j\beta_j}, \quad j = \{A, B\}, \quad \ell = 1, \ldots, n_{ij} \tag{2}$$

where $\Lambda_{0\ell j}$ is the estimated cumulative, or integrated, hazard associated with the ith individual in the ℓth time interval from a piecewise constant exponential hazard model fit to group j. And n_{ij} refers to the number of episodes of risk contributed by individual i in group j.

- We can view $\Lambda_{0\ell j}$ as the expected number of events experienced by the ith individual in the ℓth interval of exposure to risk, assuming a time-homogeneous Poisson process with rate $\lambda_{i\ell j}$ that is observed until either a first event occurs or the sub-interval of time has elapsed without the event occurring (see, e.g., Aitkin and Clayton 1980, Barlow and Proschan 1975, Holford 1980).[10]

For the (piecewise constant) exponential model, the total number of events in group j equals the sum of the estimated integrated hazards. That is,

$$\sum_{i=1}^{N_j} d_{ij} = \sum_{i=1}^{N_j}\sum_{\ell=1}^{n_{ij}} \Lambda_{0\ell j}e^{X_j\beta_j}, \quad j = \{A, B\} \tag{3}$$

- Each individual contributes $\Delta t_{i\ell}$ units of exposure to the ℓth time interval. It follows that an individual's total exposure equals the individual's event or censoring age,

$$\sum_{\ell=1}^{n_{ij}} \Delta t_{i\ell j} = t_{ij}.$$

- The number and the widths of the age intervals are chosen exogenously and must be the same for each group. Specifically, we define a set of cut-points, $[\tau_0, \tau_1), [\tau_0, \tau_1), \ldots, [\tau_\ell, \infty)$, or pieces, for the piecewise constant model that are common to both groups. Then, given an individual's event or censoring time t_i, we determine an individual's exposure in the ℓth interval as,

$$\Delta t_{i\ell} = \begin{cases} 0 & \text{if } \; t_i < \tau_{\ell-1}, \\ t_i - \tau_{\ell-1} & \text{if } \; \tau_{\ell-1} \leq \tau_\ell, \\ \tau_\ell - \tau_{\ell-1} & \text{if } \; t_i > \tau_\ell. \end{cases}$$

- This results in n_i sub-episodes of risk for individual i.
  Note that the sum of the subinterval exposures over all individuals necessarily equals the total exposure in the sample under consideration, i.e.,

$$\sum_{i=1}^{N_j}\sum_{\ell=1}^{n_{ij}} \Delta t_{i\ell j} = \sum_{i=1}^{N_j} t_{ij}, \quad \text{for } \; j = \{A, B\}.$$

Combining this with Eq.(3), we can show the equivalence between the exposure-averaged predicted event counts and the observed rates,

$$r_j = \frac{\sum_{i=1}^{N_j} d_{ij}}{\sum_{i=1}^{N_j} t_{ij}} = \frac{\sum_{i=1}^{N_j}\sum_{\ell=1}^{n_{ij}} \Lambda_{0\ell j}e^{X_j\beta_j}}{\sum_{i=1}^{N_j}\sum_{\ell=1}^{n_{ij}} \Delta t_{ij\ell}}, \quad \text{for } j = \{A, B\}$$

[10] Aitkin, M., and D. Clayton. 1980. "The Fitting of Exponential, Weibull, and Extreme Value Distributions to Complex Survival Data using GLIM." *Applied Statistics* 29:156–63; Barlow, R.E., and F. Proschan. 1975. *Statistical Theory of Reliability and Life Testing.* New York: Holt, Rinehart and Winston; Holford, T. 1980. "The Analysis of Rates and of Survivorship Using Log-Linear Models." *Biometrics* 36:299–305.

Example: The illustrative example considers racial differences in early family formation rates using 1st premarital births drawn from the NLSY79. A plot of the survivor function and cumulative hazard of a first premarital birth for White and Black women in the NLSY79 cohort is shown in Figure 1. The ratios of the cumulative hazards at age t reflect the hazard ratios at age t. We adopt a proportional hazards model specification with piecewise constant hazards over 6 age intervals, [12,16), [16,18), [18,20), [20,22), [22,24), [24+), which allows time dependence in the baseline hazard by age and is similar to the partitioning used in past research in this area.

It is convenient to parameterize the baseline log hazard separately from the structural part of the model by excluding the conventional intercept and including a set of dummy variables for the 6 age intervals, $D_{i1j}, \ldots, D_{i6j}$ and a corresponding set of parameters for the log baseline hazard, $a_{1j}, \ldots, a_{6j}$, which results in the following model specification for the rate:

$$\lambda_{i\ell j} = \exp(a_{1j} D_{i1j} + a_{2j} D_{i2j} + \ldots + a_{6j} D_{i6j} + \mathbf{z}'_{i\ell j} \boldsymbol{\gamma}_j) = e^{a_{\ell j} + \mathbf{z}'_{i\ell j} \boldsymbol{\gamma}_j}.$$

Assuming a constant exponential hazard for each piece, the integrated hazard can be written as $\Lambda_{i\ell j} = \Delta t_{i\ell j} \lambda_{i\ell j}$. Using the so called "Poisson Trick," we exploit the similarity between the loglinear model for counts and the exponential model for rates by including the logged exposure to risk in the ℓth interval ($\log \Delta t_{i\ell}$) as an "offset" term in a Poisson regression model. This approach yields a piecewise constant exponential hazard rate model (see, e.g., T. Holford 1976, Laird, N. and D. Oliver 1981).[11] Eq.(2) can now be written as,

$$F(X_j \beta_j) = e^{a_{\ell j} + \mathbf{z}'_{i\ell j} \boldsymbol{\gamma}_j + \log \Delta t_{i\ell j}}, \quad j = \{A, B\}.$$

More details on the decomposition are given in Appendix A.

Preliminary Analysis

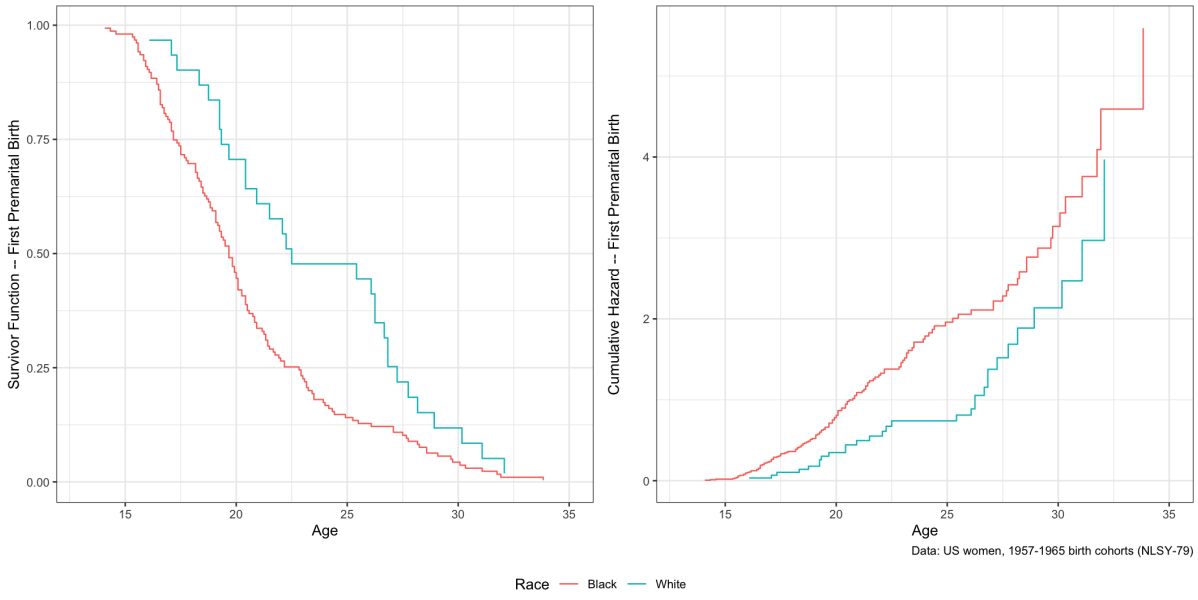


Figure 1: Survivor Function and Cumulative Hazard of First Premarital Birth by Race

We use `R` functions from the hazard rate decomposition repository on GitHub. A hazard rate decomposition using the `hazdcmp Stata` module available from the `ssc` repository is shown in Appendix B.

[11] Laird, N. and Olivier, D. (1981) "Covariance Analysis of Censored Survival Data Using Log-Linear Analysis Techniques.", *Journal of the American Statistical Association* 76, 231-240. Holford, T.R. (1976) "Life Tables with Concomitant Information." *Biometrics* 32, 587-597.

R set-up

A first step is to determine if there are meaningful differences between the two groups on outcome measures and group-specific attributes.[12] Below is a table of descriptives. We use `describe` in the `psych` library and `kable` to produce a table grouped by race. This also provides raw material for making a graphical display of group differences in attributes.

```
# read wide data for descriptives by race

df    <- read.dta(file="hazdat_wide.dta")
dfsel <- select(df, -c(iid, famid)) -> dfsel
lsdesc <- describe(dfsel ~ blk, skew=F, ranges=F, fast=T)
  lsdesc[[1]] %>% dplyr::select(-c(vars)) -> lsW
  lsdesc[[2]] %>% dplyr::select(-c(vars)) -> lsB
  cn <- colnames(lsW)
  row.omit <- c("blk")
  bind_cols(lsW[!(row.names(lsW) %in% row.omit),],
            lsB[!(row.names(lsW) %in% row.omit),]) %>%
  kable(format="latex", booktabs=T, digits=2, linesep="",
        col.names=rep(cn,2)) %>%
  add_header_above(c(" "=1, "White"=4, "Black"=4)) %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 1: Descriptive Statistics by Race

	White				Black			
	n	mean	sd	se	n	mean	sd	se
medu	2287	11.87	2.41	0.05	1357	10.76	2.59	0.07
nosibs	2287	1.90	1.75	0.04	1357	2.72	2.56	0.07
pctsmom	2287	0.06	0.17	0.00	1357	0.24	0.35	0.01
nfamtran	2287	0.49	0.95	0.02	1357	0.62	0.92	0.03
magebir	2287	25.48	5.99	0.13	1357	24.91	6.77	0.18
agepmb	2287	23.24	4.89	0.10	1357	22.10	5.28	0.14
pmb	2287	0.11	0.32	0.01	1357	0.56	0.50	0.01
inc1000	2287	1.00	0.62	0.01	1357	0.55	0.42	0.01

```
# set-up plot for B-W differences in means
lsWa <- lsW[!(row.names(lsW) %in% row.omit),]
lsBa <- lsB[!(row.names(lsW) %in% row.omit),]

dfdif <- function(df1, df2) {
    alpha = .05
    z.alpha = qnorm(alpha/2, lower.tail=F)
    dm  = df1$mean - df2$mean
    dsedm = sqrt(df1$se^2 + df2$se^2)
    cidU  = dm + z.alpha*dsedm
    cidL  = dm - z.alpha*dsedm
    term  = rownames(df1)
return(data.frame(term =term, dm=dm, lowerCL = cidL, upperCL = cidU))
}
```

[12] If a difference is not statistically discernable, then it is not worth decomposing.

```
diffm <- dfdif(lsBa, lsWa)
  ggplot(diffm, aes(x=dm, y=factor(term, levels=unique(term)))) +
    geom_point(color="#ec9db5")  +
    geom_errorbar(aes(xmin = lowerCL,
                      xmax = upperCL,
                     width =.35, color="#ec9db5"),
                 linewidth = 1,
                show.legend = F) +
  theme(axis.text = element_text(size = 12)) +
  geom_vline(xintercept=0, col="orange", linetype=3, linewidth=1.2) +
  labs(x="Black-White Difference in Means", y="")
```

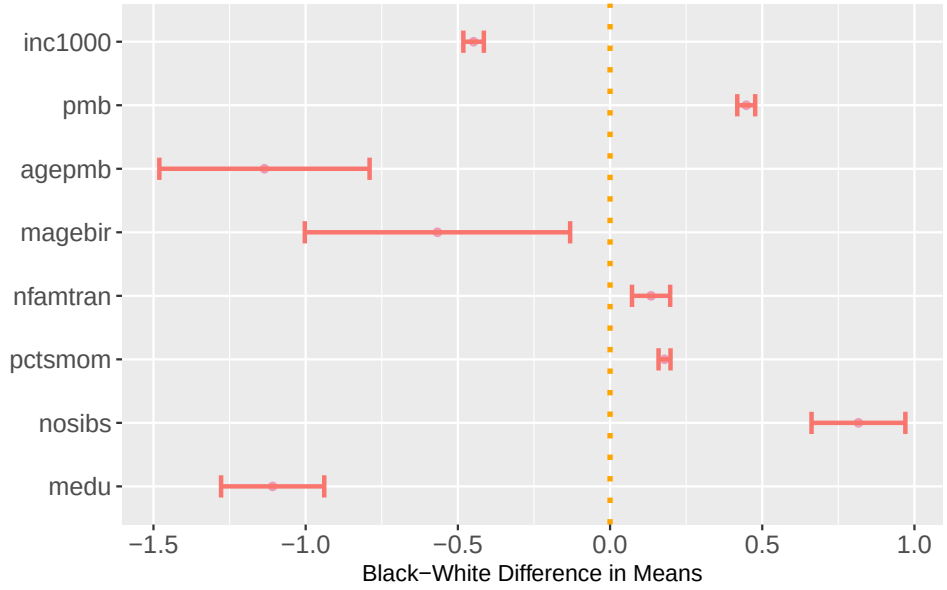


Figure 2: Black-White Difference in Covariate Means

It is helpful to circle back to the descriptive results in Table 1 and Figure 2 when discussing the detailed contributions to the E component of the decomposition results. The results above show significant differences in means on a number of family background characteristics that include:

- R mother's education (`meduc`)
- number of older siblings (`nosibs`)
- percent of time R lived in a single mother household from birth to time of event or age 18 (if censored) (`pctsmom`)
- mother's age when R was born (`magebir`)
- family income at R age 14 (`inc1000`) in $10,000 adjusted for family size.

We also see differences in the premarital birth outcomes (`pmb` and `agepmb`).[13]

The empirical first premarital birth rate is shown to be 4.82 per 1,000 per year for Whites compared to 25.3 per 1,000 per year for Blacks, with a difference of 20.5 first premarital births per 1,000 per year, and a rate ratio of 5.25.

```
df %>%
  group_by(blk) %>%
  summarize(rate = 1000 * sum(pmb)/sum(agepmb)) %>%
```

[13]The mean is not appropriate for a right censored variable such as `agepmb`.

```
  mutate(drate = rate - rate[1],
         hratio = rate/rate[1])
```

```
# A tibble: 2 x 4
    blk  rate drate hratio
  <dbl> <dbl> <dbl>  <dbl>
1     0  4.82   0      1
2     1 25.3   20.5   5.25
```

Set up Survival Data

Construct split episode data using age at first premarital birth (`agepmb`) and the event/censoring indicator (`pmb`). Construct the exposure variable (`Dt`), factor labels for time interval (`Int`) and race. Then display summary information on age-specific premarital birth rates by race in a cursory table with a plot to show the disparity (scaled to rates per 1,000 per year of age).

```
t.u  <- c(16, 18, 20, 22, 24) # cutpoints
 dfsplit <-survSplit(Surv(agepmb, pmb) ~., df,
                       cut = t.u,
                       end = "agepmb",
                     event = "pmb",
                     start = "start",
                   episode = "Int")
```

```
  dfsplit <- mutate(dfsplit,
            Int=factor(Int,
            labels=c("[12,16)", "[16,18)", "[18,20)",
                     "[20,22)", "[22,24)", "[24,36)")),
            Race=factor(blk,
            labels = c("White", "Black")),
                Dt = agepmb - start)

 vInt <- with(dfsplit, unique(as.numeric(Int)))
 lInt <- with(dfsplit, levels(Int))
 dfsplit %>% group_by(Int, Race) %>%
   summarize(nevent = sum(pmb),
             pyrisk = sum(Dt),
              nrisk = n(),
               Rate = nevent/pyrisk * 1000) -> tabdat

 tabdat %>% filter(Race=="Black") %>% ungroup() %>%
       dplyr::select(-c(Race,Int)) -> tabB
 tabdat %>% filter(Race=="White") %>% ungroup() %>%
       dplyr::select(-c("Race")) -> tabW
 dfHR <- data.frame(HR = with(tabB, Rate)/with(tabW, Rate))
 cnB  <- colnames(tabB)
 cnW  <- colnames(tabW)
 cnHR <- colnames(dfHR)
 bind_cols(tabW, tabB, dfHR) %>%
   kable(format="latex", booktabs=T,
         digits=2, linesep="",
         col.names=c(cnW, cnB, cnHR)) %>%
   add_header_above(c(" "=1, "White"=4, "Black"=4, " "=1)) %>%
   kable_styling(latex_options = "HOLD_position")
```

Table 2: Age-specific Premarital Birth Rates by Race

	White				Black				
Int	nevent	pyrisk	nrisk	Rate	nevent	pyrisk	nrisk	Rate	HR
[12,16)	17	36556.69	2287	0.47	76	21640.61	1357	3.51	7.55
[16,18)	56	4287.55	2232	13.06	204	2312.13	1269	88.23	6.76
[18,20)	59	3575.63	2012	16.50	190	1794.20	1025	105.90	6.42
[20,22)	47	2754.55	1559	17.06	134	1264.13	754	106.00	6.21
[22,24)	20	2040.25	1181	9.80	70	925.68	534	75.62	7.71
[24,36)	57	3934.38	860	14.49	85	2058.64	386	41.29	2.85

```
ggplot(tabdat, aes(as.numeric(Int), Rate, col=Race)) +
    geom_smooth() + scale_x_continuous(breaks=vInt, labels=lInt) +
    labs(x = c("Age Interval"),
         title="Age-Specific Premarital Birth Rates by Race (per 1,000/yr)") -> dp1
```

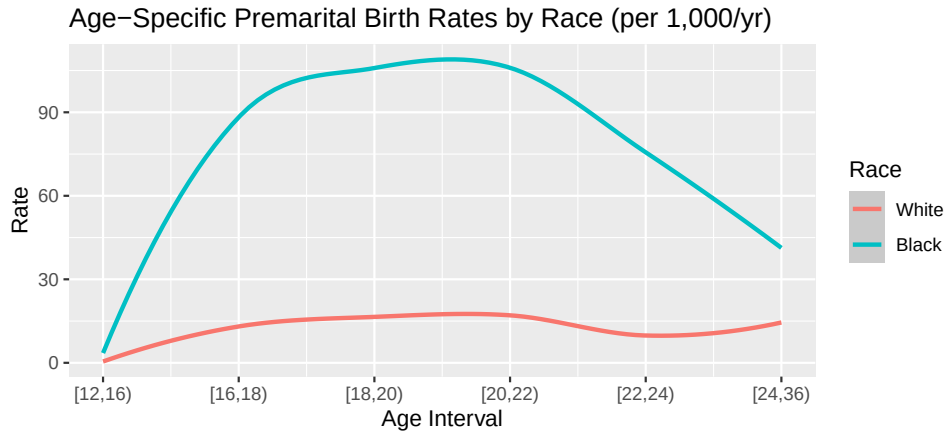


Figure 3: Rates by Race

The `HR` column in Table 2 shows that the risk of a first premarital for Black woman relative to White women is higher by a factor of 2.85 (at ages 24 and older) to 7.71 between the ages of 20 to 24. This is borne out by Figure 3.

Rate Models

With the data arranged in split-episode format, we fit piecewise constant hazard models by race using `modelsummary` to show results. It is worthwhile to check agreement in rates between the wide (unsplit) data and the long (split-episode data).

The coefficient plot (shown in Figure 4) is more reader friendly than Table 3, and is wrangled in `broom`. We will circle back to these coefficients when discussing the detailed contributions to the C component of the difference in rates.

```
# models
mW <- glm(pmb ~ Int + pctsmom + nfamtran + medu +
    inc1000 + nosibs + magebir + offset(log(Dt)) - 1,
    sub=Race=="White", family="poisson", data=dfsplit)

mB <- glm(pmb ~ Int + pctsmom + nfamtran + medu +
```

Table 3: Hazard and Hazard Ratios of 1st Premarital Birth by Race

	White		Black	
	Est.	[conf.low, conf.high]	Est.	[conf.low, conf.high]
Int[12,16)	0.006	[0.002, 0.014]	0.014	[0.009, 0.023]
Int[16,18)	0.175	[0.078, 0.385]	0.368	[0.231, 0.582]
Int[18,20)	0.244	[0.109, 0.542]	0.455	[0.285, 0.725]
Int[20,22)	0.276	[0.120, 0.624]	0.473	[0.292, 0.762]
Int[22,24)	0.165	[0.067, 0.397]	0.348	[0.208, 0.577]
Int[24,36)	0.245	[0.108, 0.549]	0.194	[0.117, 0.319]
pctsmom	2.167	[1.118, 3.953]	1.117	[0.901, 1.377]
nfamtran	1.336	[1.207, 1.465]	1.162	[1.074, 1.253]
medu	0.873	[0.834, 0.916]	0.937	[0.910, 0.965]
inc1000	0.658	[0.501, 0.851]	0.570	[0.459, 0.703]
nosibs	1.182	[1.099, 1.267]	1.053	[1.018, 1.089]
magebir	0.949	[0.925, 0.973]	0.971	[0.959, 0.984]
Num.Obs.	10 131		5325	
Log.Lik.	−1270.662		−2667.725	

```
    inc1000 + nosibs + magebir + offset(log(Dt)) - 1,
    sub=Race=="Black", family="poisson", data=dfsplit)
```

```
modelsummary(list(White=mW, Black=mB), exponentiate=T,
statistic=c("[{conf.low}, {conf.high}]"),
shape = term ~ model + statistic,
title = "\\label{tab:tab3} Hazard and Hazard Ratios of
1st Premarital Birth by Race", escape=F, output="latex",
gof_omit=c("AIC|BIC|RMSE"))
```

```
bind_rows(tmW, tmB) %>%
  mutate(Race = factor(Race,
       labels = c("White", "Black"))) %>%
  ggplot(aes(estimate, factor(term, levels=unique(term)), col=Race)) +
  geom_point(position=position_dodge(width=1) ) +
  geom_errorbar(aes(xmin=conf.low,
                    xmax=conf.high,
                    y=factor(term, levels=unique(term))),
                    width=.35, position=position_dodge(width=1)) +
  geom_vline(xintercept=1, col="orange4", linetype=2) +
  labs(x="Hazards/Hazard Ratios---1st Premarital Birth by Race", y="") -> dp2B
```

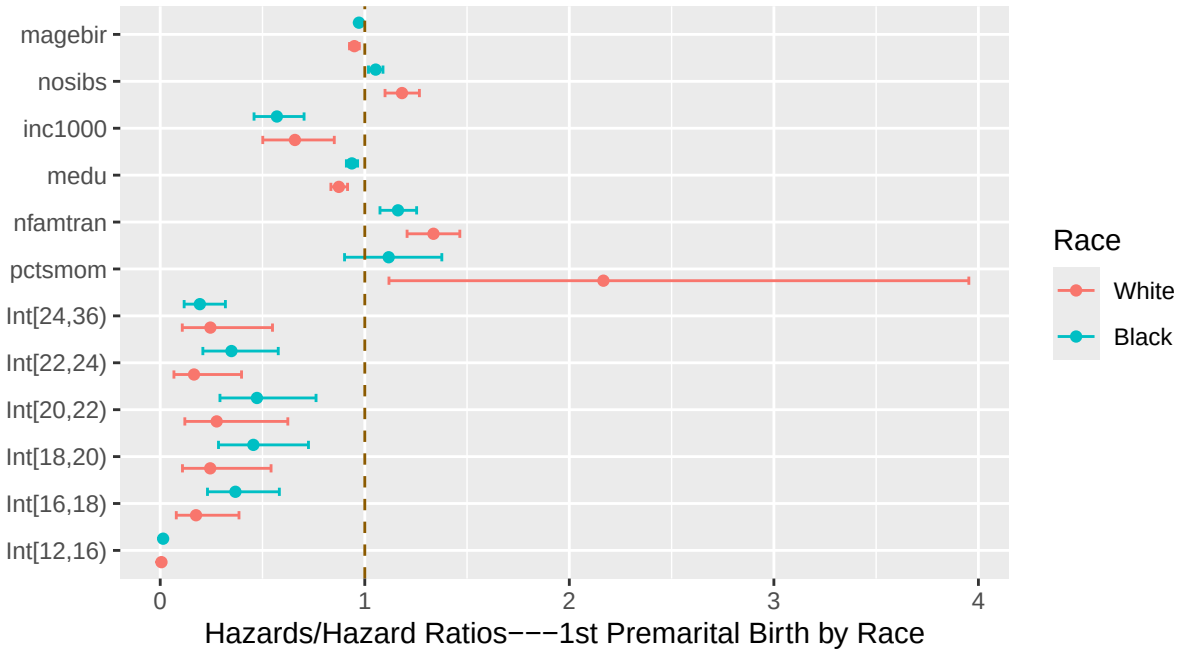


Figure 4: Coefficient Plot

Decomposition Set-up

We first consider a decomposition of a difference in rates from the perspective of the high outcome group. In this case, Black women experience higher rates of a first premarital birth and are considered as group A. The companion paper applies an additional step that averages two decompositions carried out from each group's perspective. We will discuss that strategy later on.

The workhorse of the decomposition routine is `svyglm` in the `survey` library. So, we need to attend to the specifics of setting up the design. For a generic analysis, declare an `id` uniquely identifying subjects and unit `weights`. The `id` is needed to ensure that person-level covariate means are calculated. Other possibilities would include defining `clusters` based on `famid` to account for dependencies of outcomes within families, or other aspects of the survey design. We must define the two groups as subsets, `Asub` and `Bsub`, of the survey weighted data, and pass these to the model.

```
dfsvy <- svydesign(ids=~iid, weights=~1, nest=TRUE, data=dfsplit)
Asub  <- subset(dfsvy, Race == "Black")
Bsub  <- subset(dfsvy, Race == "White")

# fit standard decomposition (group A as reference)

m1a <- decomp.pwcexp(pmb ~ Int + pctsmom + nfamtran + medu +
            inc1000 + nosibs + magebir + offset(log(Dt)) - 1,
          Asub, Bsub, scale=1000, printit=FALSE)

# fit reversed decomposition (group B as reference)
m1b <- decomp.pwcexp(pmb ~ Int + pctsmom + nfamtran + medu +
            inc1000 + nosibs + magebir + offset(log(Dt)) - 1,
          Asub, Bsub, scale=1000, reverse=TRUE, printit=FALSE)
```

Accessing the decomposition routine requires `sourcing` the function script downloaded from `GitHub` as follows.

```
source("~/yourpath/hazard_decomp_functions_svy.R")
```

The call to `decomp.pwcexp` will fit piecewise constant rate models to each group defined by `Asub` and `Bsub` using `svyglm`. The model standard errors will be "robust" as the `quasipoisson` family is used for the Poisson family by `svyglm` by default.[14] Model results are processed to get the detailed decomposition components and their asymptotic variances. The delta method is used to build the covariance matrices of the decomposition components as nonlinear transformations of the model covariance matrices. Thus, the variances of the decomposition components inherit the properties of the survey design specified by the user. These details are discussed in Appendix A.

If the `printit` option is `TRUE`, output is displayed at the terminal. Otherwise, it is hidden. Here we use helper functions to extract and process the components of the `decomp.pwc` object as shown below using `get.decomp` and placing results in the objects `E.m1a` and `C.m1a`.

```
E.m1a <- get.decomp(m1a, scale=1000, component = "E")
C.m1a <- get.decomp(m1a, scale=1000, component = "C")
```

These results get passed to a helper function to produce the decomposition results in Table 4.

```
# E & C in that order
dfT <- tabFun(E.m1a, C.m1a) # returns list with E [[1]] and C [[2]]
cn  <- colnames(dfT[[1]])
lrn <- length(rownames(dfT[[1]])) - 1

bind_cols(dfT[[1]], dfT[[2]])  %>%
  kable(format="latex", booktabs=T, linesep="",
        caption="Decomposition Components",
        col.names=rep(cn,2), digits=3) %>%
  add_header_above(c(" "=1, "Characteristics (E)"=4,
                     "Coefficients (C)"=4)) %>%
  row_spec(lrn, hline_after = TRUE, extra_latex_after = "%") %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 4: Decomposition Components

	Characteristics (E)				Coefficients (C)			
	b	b.lower	b.upper	pct	b	b.lower	b.upper	pct
Int[12,16)	-7.299	-9.293	-5.306	-35.628	2.065	-0.420	4.551	10.081
Int[16,18)	-0.565	-0.884	-0.246	-2.757	1.594	-0.540	3.729	7.781
Int[18,20)	0.483	0.155	0.810	2.355	1.075	-0.669	2.820	5.249
Int[20,22)	0.471	0.132	0.810	2.299	0.620	-0.565	1.805	3.027
Int[22,24)	0.640	0.278	1.002	3.123	0.590	-0.280	1.460	2.880
Int[24,36)	0.706	0.421	0.991	3.446	-0.129	-0.674	0.416	-0.630
pctsmom	0.558	-0.559	1.674	2.722	-0.346	-0.719	0.027	-1.687
nfamtran	0.571	0.233	0.909	2.786	-0.592	-1.106	-0.079	-2.891
medu	2.038	0.957	3.119	9.948	7.270	1.558	12.981	35.483
inc1000	7.093	5.324	8.863	34.624	-1.249	-4.668	2.169	-6.097
nosibs	1.192	0.384	1.999	5.816	-1.915	-3.215	-0.614	-9.345
magebir	0.464	0.238	0.690	2.263	5.153	-1.279	11.584	25.152
Total	6.350	4.643	8.058	30.997	14.137	11.979	16.294	69.003

We can `rbind` the results and construct a suitable format for `ggplot` shown in Figure 5.

[14] It would be possible to change this option with some additional programming, but it is harmless to leave it as is.

```r
# keep rows as needed
rowkeep   <- nrow(E.m1a) - 2

E.m1a <-  mutate(E.m1a, type=1) %>% filter(row_number() < rowkeep)
C.m1a <-  mutate(C.m1a, type=2) %>% filter(row_number() < rowkeep)

bind_rows(E.m1a, C.m1a) %>%
  mutate(type = factor(type,
       labels = c("Characteristic Component", "Coefficient Component"))) %>%
  ggplot(aes(b, factor(term, levels=unique(term)), col=type)) +
  facet_wrap(~type, scales="free_x") +
  geom_point(show.legend=F) +
  geom_errorbar(aes(xmin=b.lower,
                    xmax=b.upper,
                    y=factor(term, levels=unique(term))),
                    show.legend=F, width=.35) +
  geom_vline(xintercept=0, col="orange", linetype=2) +
  labs(x="Decomposition Components", y="") -> dp3a
```

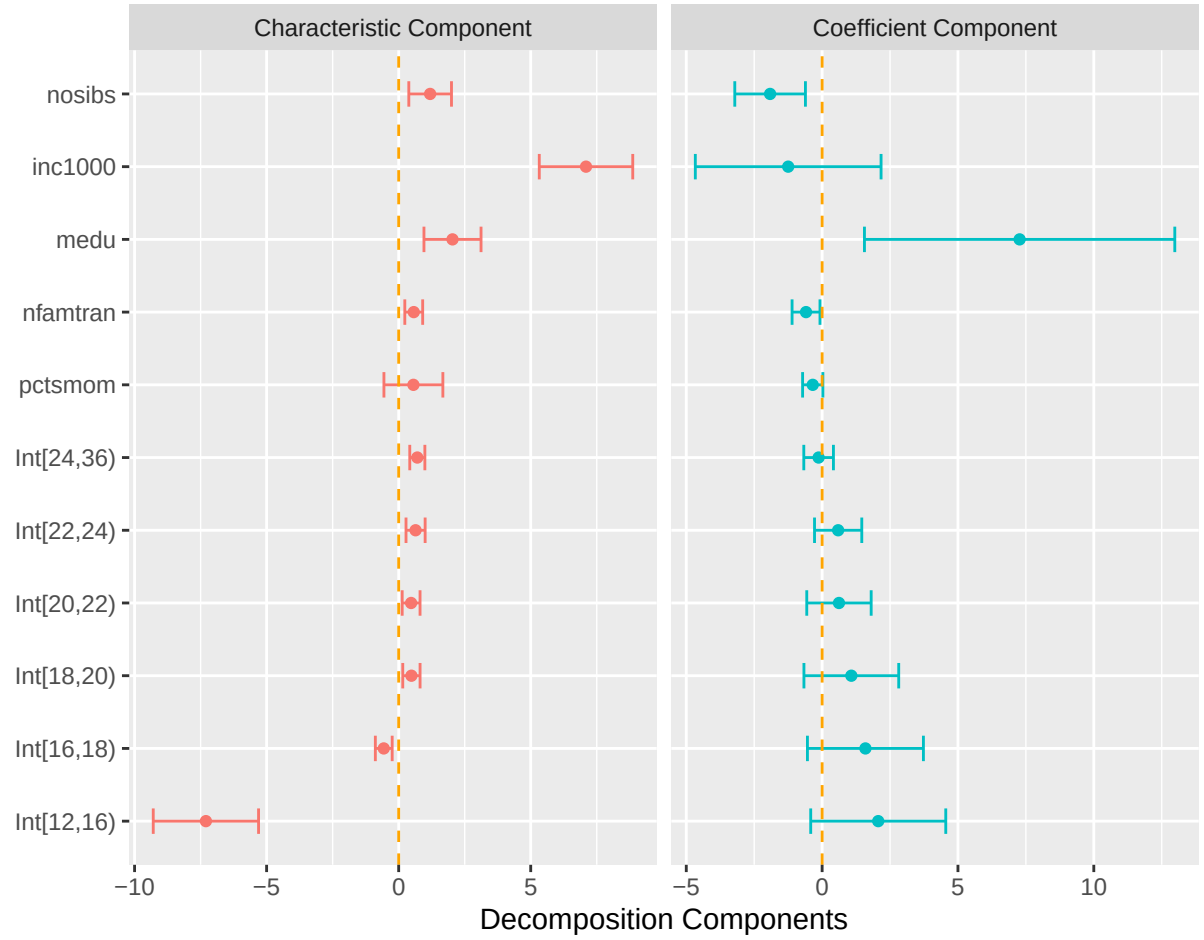


Figure 5: Coefficient Plot

Interpreting Results: We see from Table 4 that the E component accounts for about 31% of the overall Black-White difference in 1st premarital birth rates. We should note that in the normalization adopted here,

a positive E or C coefficient reflects the (scaled) amount that the gap would be expected to *decrease* if we could equalize characteristics and coefficients across groups. Negative signed E and C imply that the gap would *increase*. We can view the coefficient for overall E of 6.35 as the amount by which the racial premarital birth gap would decrease if Black women's observed characteristics matched those of White women, while retaining their specific returns to risk on these characteristics (about 6 births per 1,000). In particular, making Black family income equal to that of Whites is expected to reduce the gap by about 7 births per 1,000 (or about 35%). Equalizing maternal education would be expected to reduce the premarital birth rate gap by 2 births per 1,000 (or about 10%). It is helpful (and often necessary) to refer back to the descriptive statistics to make sense of decomposition results. For example, Table 3 shows that Black family income is about half that of Whites in the NLSY79 and maternal educational levels differ by more than one year.

The C component accounts for 69% of the overall Black-White premarital birth rate gap. The coefficient for overall C implies a 14.14 births per 1,000 reduction in the gap if Black returns to risk equaled those of Whites, and Black characteristics were fixed to White's levels. For example, if Black respondents experienced the same returns to maternal education as White respondents, the premarital birth gap would be expected to shrink by about 35.5 percent (or 7.3 births per 1,000). It is helpful to consult Table 3 that shows the stronger association between maternal education and risk of premarital birth among Whites (i.e., a one-year increase in maternal schooling reduces the hazard of a 1st premarital birth by about 13% for Whites compared to about 6.3% for Blacks.)

The Index Number Problem and a Workaround. The results thus far have decomposed the difference in rates from the perspective of Blacks (group A), where the total differential is maintained by weighting the White rates by the Black effects in the E component and replacing the characteristics of Blacks with those of Whites in the C component. This is more easily shown using a linear model where mean substitution is appropriate.[15]

$$\begin{aligned}
\bar{Y}_A - \bar{Y}_B &= \sum_k \bar{X}_{Ak} b_{Ak} - \sum_k \bar{X}_{Bk} b_{Bk} \\
&= \left(\sum_k \bar{X}_{Ak} b_{Ak} - \sum_k \bar{X}_{Bk} b_{Ak} \right) + \left(\sum_k \bar{X}_{Bk} b_{Ak} - \sum_k \bar{X}_{Bk} b_{Bk} \right) \\
&= \sum_k (\bar{X}_{Ak} - \bar{X}_{Bk}) b_{Ak} + \sum_k \bar{X}_B (b_{Ak} - b_{Bk})
\end{aligned}$$

Thus, $E^a = \sum_k (\bar{X}_{Ak} - \bar{X}_{Bk}) b_{Ak}$ and $C^a = \sum_k \bar{X}_B (b_{Ak} - b_{Bk})$.

Similarly, a decomposition of the $A - B$ difference from group B's perspective is

$$\begin{aligned}
\bar{Y}_A - \bar{Y}_B &= \sum_k \bar{X}_{Ak} b_{Ak} - \sum_k \bar{X}_{Bk} b_{Bk} \\
&= \left(\sum_k \bar{X}_{Ak} b_{Bk} - \sum_k \bar{X}_{Bk} b_{Bk} \right) + \left(\sum_k \bar{X}_{Ak} b_{Ak} - \sum_k \bar{X}_{Ak} b_{Bk} \right) \\
&= \sum_k (\bar{X}_{Ak} - \bar{X}_{Bk}) b_{Bk} + \sum_k \bar{X}_A (b_{Ak} - b_{Bk})
\end{aligned}$$

Thus, $E^b = \sum_k (\bar{X}_{Ak} - \bar{X}_{Bk}) b_{Bk}$ and $C^b = \sum_k \bar{X}_A (b_{Ak} - b_{Bk})$. And, $E^a \neq E^b$ if $b_{Bk} \neq b_{Ak}$. And, $C^a \neq C^b$ if $\bar{X}_{Bk} \neq \bar{X}_{Ak}$.

[15] The substutution approach does not apply to nonlinear models. In this case, we can average the results from two decompositions.

It can be shown that the $\bar{Y}_A - \bar{Y}_B$ decomposition obtained by swapping b_{Ak} with b_{Bk} and $\bar{X}_{Bk}$ with $\bar{X}_{Ak}$ is a $\bar{Y}_B - \bar{Y}_A$ decomposition with B as the reference group. This, of course, results in a change in sign of the difference as well as the components of the difference. From a practical standpoint we can average the detailed E and C components from an $A - B$ decomposition and a $B - A$ decomposition, while maintaining the $A - B$ difference. This gives the following results,

$$\bar{Y}_A - \bar{Y}_B = \sum_k (\bar{X}_{Ak} - \bar{X}_{Bk}) b_k + \sum_k \bar{X}_k (b_{Ak} - b_{Bk}),$$

where $b_k = (b_{Ak} - b_{Bk})/2$ and $\bar{X}_k = (\bar{X}_{Ak} - \bar{X}_{Bk})/2$.

Rather than using group A's or group B's coefficients in E, we use the average of A and B coefficients (i.e., from a standard and reversed decomposition). Similarly, the C component reflects the average of the characteristics from both groups.

Averaging two Decompositions

The companion paper presents results from averaging the results from a standard and reversed decomposition. Here we use a helper function `ave.decomp`. Averaging permits an alternative assessment of the results as they are less dependent on the choice of the reference group. This is akin to using the pooled population as the standard, and is a compromise to get around the index problem.

```
E.ave <- ave.decomp(m1a, m1b, scale=1000, component="E")
C.ave <- ave.decomp(m1a, m1b, scale=1000, component="C")
```

These results get passed to a helper function to produce the averaged decomposition results in Table 5.

```
# E & C in that order
dfT <- tabFun(E.ave, C.ave) # returns list with E [[1]] and C [[2]]
cn  <- colnames(dfT[[1]])
lrn <- length(rownames(dfT[[1]])) - 1

bind_cols(dfT[[1]], dfT[[2]])  %>%
  kable(format="latex", booktabs=T, linesep="",
        caption="Decomposition Components",
        col.names=rep(cn,2), digits=3) %>%
  add_header_above(c(" "=1, "Characteristics (E)"=4,
                     "Coefficients (C)"=4)) %>%
  row_spec(lrn, hline_after = TRUE, extra_latex_after = "%") %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 5: Decomposition Components

	Characteristics (E)				Coefficients (C)			
	b	b.lower	b.upper	pct	b	b.lower	b.upper	pct
Int[12,16)	-5.021	-6.430	-3.611	-24.507	2.752	0.357	5.147	13.431
Int[16,18)	-0.436	-0.661	-0.210	-2.127	1.968	0.071	3.865	9.607
Int[18,20)	0.376	0.144	0.607	1.835	1.188	-0.182	2.558	5.799
Int[20,22)	0.362	0.122	0.601	1.765	0.660	-0.234	1.554	3.222
Int[22,24)	0.490	0.234	0.746	2.392	0.601	-0.026	1.228	2.932
Int[24,36)	0.447	0.245	0.649	2.183	-0.131	-0.521	0.260	-0.639
pctsmom	0.886	0.097	1.676	4.326	-1.110	-2.121	-0.099	-5.419
nfamtran	0.457	0.218	0.696	2.229	-0.810	-1.327	-0.293	-3.952
medu	1.678	0.914	2.443	8.191	8.112	3.606	12.618	39.597
inc1000	4.369	3.118	5.620	21.325	-1.091	-3.222	1.040	-5.326
nosibs	1.195	0.623	1.766	5.831	-2.816	-4.205	-1.427	-13.745
magebir	0.362	0.202	0.522	1.766	5.999	0.663	11.335	29.283
Total	5.165	3.834	6.495	25.209	15.323	13.508	17.137	74.791

We can `rbind` the results of averaging and construct a suitable format for `ggplot` shown in Figure 6.

```
# keep rows as needed
rowkeep   <- nrow(E.ave) - 2

E.ave <-  mutate(E.ave, type=1) %>% filter(row_number() < rowkeep)
C.ave <-  mutate(C.ave, type=2) %>% filter(row_number() < rowkeep)

bind_rows(E.ave, C.ave) %>%
  mutate(type = factor(type,
       labels = c("Characteristic Component", "Coefficient Component"))) %>%
  ggplot(aes(b, factor(term, levels=unique(term)), col=type)) +
  facet_wrap(~type, scales="free_x") +
  geom_point(show.legend=F) +
  geom_errorbar(aes(xmin=b.lower,
                    xmax=b.upper,
                    y=factor(term, levels=unique(term))),
                    show.legend=F, width=.35) +
  geom_vline(xintercept=0, col="orange", linetype=2) +
  labs(x="Decomposition Components", y="") -> dp3b
```

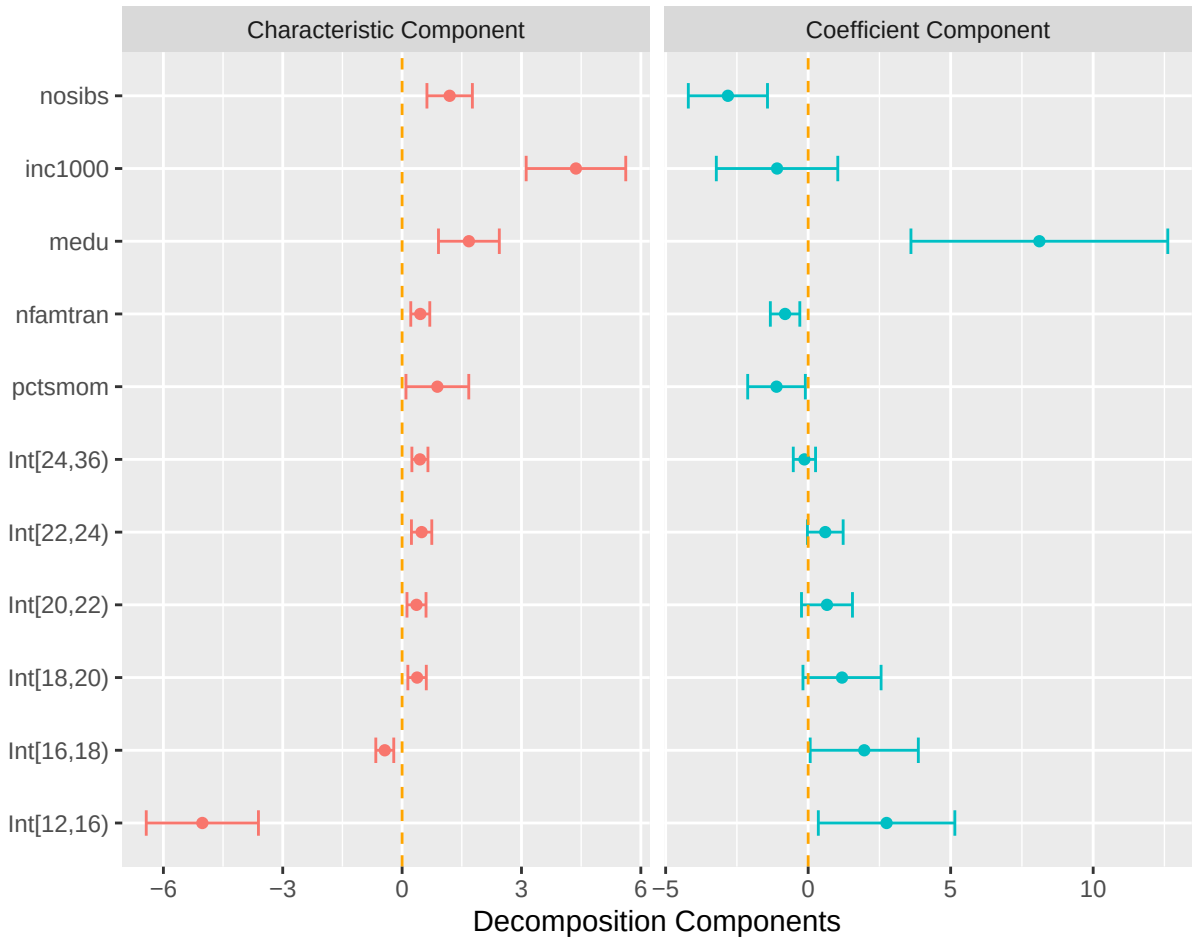


Figure 6: Coefficient Plot

Interpreting Results

The following interpretation of the coefficients holds for the two-way decomposition implemented here. The total of the respective decomposition components is $E = \sum E_k$ and $C = \sum C_k$. These sum to the total difference, $R = E + C$.

We can see from Table 5 that roughly 25% of the premarital birth gap can be attributed to racial differences in compositional factors (E); the remainder is attributable to differences in the effects of those factors (C).

Excluding the baseline hazard, main contributors to E are racial differences in family income (21%), R's mother's years of education (8%), number of older siblings (6%), and mother's age at R's birth (2%).

Notes:

The E component is the major component of interest in most settings. Note that a positive E_k coefficient quantifies how much the gap would decrease if groups were equal on characteristics, evaluated at average of the two groups' β_k values. A negative E_k coefficient implies an increase in the gap. Specific E_k (Table 5) are discussed below in relation to the proportional hazards results (Table 3) and the descriptive statistics (Table 1).

- **Family Changes**: Based on the results from the proportional hazard model in Table 3, each change in family structure (or family transition) increases the risk of a first premarital birth by 16% for Blacks and 34% for Whites.

However, the descriptive statistics in show that Whites experience fewer of these transitions on average than Blacks, with means of 0.49 and 0.62 transitions, respectively. The decomposition results in Table 5 show that this compositional disadvantage for Black women contributes 0.46 (a 2.2% reduction) to the overall racial gap in premarital birth rates.

- **Family Income**: Considering the family income effect in Table 3, we see that a $10,000 increase in adjusted family income is associated with a 34% and 43% decrease in the risk of premarital birth for Whites and Blacks, respectively. Despite similar returns to income, average income in Black families in the NLSY79 is 55% that of White families as shown in Table 1.

Table 5 shows that compositional differences in family income account for a 4.37 (per 1,000) decrease in the premarital birth gap. This represents 21% of the overall racial gap in premarital birth rates. Among the compositional factors considered here, making family incomes and number of older siblings of Blacks equal to that of Whites would produce the largest reductions in the racial gap in the premarital birth rate.

Next evaluate the C component. A similar interpretation applies to the effects component, C. A negative coefficient indicates the expected increase in the Black-White gap if Blacks experienced the same returns-to-risk as Whites, evaluated at the average of the two groups' X_k values.

- **Number of Older Siblings**: If we consider the "number of older siblings" effects reported in proportional hazards results in Table 3, each additional elder sibling is expected to increase the risk of a 1st premarital birth by 18.2% and 5.3% for White and Black women, respectively.

From Table 5, we find that the overall Black-White gap would be expected to increase by 2.82 births per 1,000 (7.7%) if Black women were penalized by the number of older siblings to the same extent as White women.

- **Mother's Education**: Similarly, a positive C-coefficient reflects the expected decrease in the black-white gap due to equalizing an effect to the white level.

For example, Whites and Blacks experience different returns to maternal education. Based on the results from the proportional hazards models in Table 3, each additional year of mother's schooling reduces the risk of a first premarital birth by 12.6% for whites and 6.3% for blacks.

The decomposition results in Table 5 show that if higher levels of maternal education benefited Blacks to the same extent as Whites, then we would expect the Black-White gap in the premarital birth rate to decrease by 8.11 births per 1,000, or 39.6% of the overall gap.

Discussion

Differences in returns to maternal education, as well as differences in the effects of maternal age at respondent's birth are the largest contributors to the overall racial gap in 1st premarital births observed in the NLSY79. If we were to consider a hypothetical policy designed to reduce the Black-White premarital birthrate gap by equalizing socioeconomic resources across groups, this is expected to lead to a larger decrease in the compositional portion of the gap than would making groups more similar in terms of family structure (i.e., the number of family transitions, proportion of years spent in a single mother family, and number of older siblings).

However, a greater share of the total differential can be attributed to differences in the effects of maternal education and mother's age at respondent's birth. So, equalizing these effects across groups would yield the greatest reduction in the Black-White premarital birth rate gap.

Appendix A: Estimation.

The procedure is entirely model-based. The only estimation taking place is the fitting of separate hazards models for the two groups. Everything, such as β's, their covariance matrix $\mathbf{\Sigma}_{\boldsymbol{\beta}}$, the X's and offsets, and complex-survey design information, are passed to functions that build the detailed decomposition components and their asymptotic covariances.

Path Dependencence and Detailed Decomposition

A key difference between decomposition of the type of model considered here (i.e., nonlinear response) and linear models, is path dependence. In this context, path dependence refers to the detailed decomposition's dependence on the order of entry of covariates into the decomposition that prevents it from producing unique results. This is a well known problem in nonlinear decomposition, with remedies that tend to be complex and data intensive.

Yun Weights. Myeong-Su Yun described a simple solution to this problem in a path-breaking paper.[16] Yun's (2004) paper provided insight about how to approach the detailed decomposition for a wide class of nonlinear response models.[17] Detailed decompositions obtained using Yun's approach are invariant to the order that variables enter the decomposition, thus providing a unique solution.

Yun presented his working paper at the national meetings of the American Economic Association, which prompted a colleague at `StataCorp` to reach out and partially fund our efforts to develop the `Stata` add-on `mvdcmp` that greatly extended multivariate decomposition to logit, probit, complementary log-log, Poisson, and negative binomial models.[18] These ideas were also implemented in the `hazdcmp` `Stata` module as well as the decomposition routines in my `GitHub` repository. The key to Yun's approach is to approximate Eq.(1) in a way that leads to two sets of weights, $W_{\Delta_{X_k}}$ and $W_{\Delta_{\beta_k}}$, $k = 1, \ldots, K$. When multiplied by the scalars E and C, respectively, these weights produce the detailed characteristics and coefficients effects E_k and C_k. Yun derived the weights using a two-step approach. First, by evaluating Eq.(1) at the means of X_A and X_B. And second, by using a 1st order Taylor expansion to linearize the approximation around $\bar{X}_A\beta_A$ and $\bar{X}_B\beta_B$ as shown below.

$$\begin{aligned} r_A - r_B &= [F(\bar{X}_A\beta_A) - F(\bar{X}_B\beta_A] + [F(\bar{X}_B\beta_A) - F(\bar{X}_B\beta_B] + R_M \\ &= [(\bar{X}_A - \bar{X}_B)]f(\bar{X}_A\beta_A) + \bar{X}_B(\beta_A - \beta_B)f(\bar{X}_B\beta_B) + R_M + R_T, \end{aligned} \tag{4}$$

where

$$f(\bar{X}_j\beta_j) = \frac{dF(\bar{X}_j\beta_j)}{d(\bar{X}_j\beta_j)},\ j = \{A, B\},$$

and R_M and R_T denote approximation residuals resulting from evaluating $F(\cdot)$ at the mean values and by using the 1st order Taylor expansion, respectively. All three terms are scalar quantities.[19]

The kth weight applied to E is

$$W_{\Delta_{X_k}} = \frac{\beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k})}{\sum_{k=1}^{K} \beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k})} \tag{5}$$

[16] Yun, M-S. 2004. "Decomposing Differences in the First Moment," *Economics Letters*, 82: 275-280.

[17] Generalized linear models are suitable candidates here, with linear models as a trivial case.

[18] Powers, DA, Yoshioka, H, and Yun, M-S 2011. "Multivariate Decomposition for Nonlinear Response Models, *Stata Journal*, vol. 11, issue 4, 556-576.

[19] The means reflect the distribution of covariates in the original data rather than the split-episode data.

and the kth weight applied to C is,

$$W_{\Delta_{\beta_k}} = \frac{\bar{X}_{B_k}(\beta_{A_k} - \beta_{B_k})}{\sum_{k=1}^{K} \bar{X}_{B_k}(\beta_{A_k} - \beta_{B_k})} \tag{6}$$

where

$$\sum_k W_{\Delta_{X_k}} = \sum_k W_{\Delta_{\beta_k}} = 1.0.$$

The composition weights $W_{\Delta_{X_k}}$ reflect the proportionate contribution of the kth covariate to the linearization of E as determined by the magnitude of the group difference in means weighted by the reference group's effect. Similarly, the coefficient weights $W_{\Delta_{\beta_k}}$ reflect covariate k's proportionate contribution to the linearization of C as determined by the magnitude of the group difference in the effects weighted by the comparison group's mean. The weights are proportional to the contributions to the decomposition of the linear predictor, in which the relative sizes of the contributions to the explained or unexplained portions of the outcome differential are equal to the relative contributions to the decomposition of the linear predictor. The weights produced in this way are invariant to change in the scale of the covariates.

The raw difference can now be expressed in terms of the overall components as a sum of weighted sums of the unique contributions.[20]

$$r_A - r_B = E + C = \sum_{k=1}^{K} W_{\Delta X_k} E + \sum_{k=1}^{K} W_{\Delta \beta_k} C = \sum_{k=1}^{K} E_k + \sum_{k=1}^{K} C_k. \tag{7}$$

Alternative Approaches. Researchers have used identical weighting/linearization approaches that predate Yun (2004).[21] Recently, alternative approaches to the detailed decomposition in nonlinear models have applied the mean value theorem to linearize the composition component E (see e.g., Bazen et al. (2017), Kaiser (2015), Schwiebert (2015)).[22] The concern arises from Jensen's Inequality, $\overline{F(X\beta)} \neq F(\bar{x}\beta)$. Bazen et al. argue that the basis for decomposition should be based on model predictions evaluated at the sample means (i.e., $F(\bar{X}\beta)$), rather than the mean of the predictions (i.e., $\overline{F(X\beta)}$). They suggest a simple form for the detailed decomposition when based on a linear index, $F(\cdot) = F(X\beta)$, as is the case with generalized linear models. Both, Schwiebert and Bazen et al. suggest differences in characteristics should be evaluated at specific marginal effects that follow from applying the mean value theorem.[23] Bazen, et al. propose the following form for the detailed composition component (in the form of marginal effects),

$$E_k^{\mathrm{MV}} = c_{A0}\beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k}),$$

where, the scalar c_{A0} is

[20]It should be noted that results using this weighting should agree with the more intensive methods. As noted by Yun (2004, p. 279) in his description of a leading approach at that time, "the contributions of a single variable can be calculated by replacing the value of group A with that of group B sequentially one by one. Using this sequential replacement, the characteristics and coefficients effects of individual variables can be calculated no matter what kind of functional form is employed. However, the sequential replacement method is not only tedious but also problematic since it may be sensitive to the order of switching."

[21]W.E. Even and D.A. Macpherson (1993) "The Decline of Private-Sector Unionism and the Gender Wage Gap," *The Journal of Human Resources*, 28:279-296 and H. Nielson (1998) "Discrimination and Detailed Decomposition in a Logit Model," *Economics Letters*, 61: 115-120. The former does not explain how the weights were derived and considers only the detailed composition component. The latter derives Yun-type weights for the unexplained (discrimination) component using total differentiation of C with respect to a coefficient δ. As noted by Yun (2004), the choice of δ is arbitrary, and if set to the same value in each group, produces weights for C that are identical to Yun's weights.

[22]Bazen, S., et al. (2017) "An Oaxaca Decomposition for Nonlinear Models," *Journal of Economic and Social Measurement*, 42:101-121. https://doi.org/10.3233/JEM-17. Kaiser, B (2015) "Detailed Decompositions in Nonlinear Models," *Applied Economics Letters*, 22:25-29. Schwiebert, J. (2015) "A Detailed Decomposition for Nonlinear Econometric Models," *Journal of Economic Inequality (2015) 13:53–67 doi 10.1007/s10888-014-9291-x.*

[23]The mean-value approach is equivalent to a 1st order Taylor linearization. The aforementioned papers provide a clear justification of this approach. Bazen claims "there is no remainder." (Pp. 120). Regarding the detailed decomposition. "... in general this cannot be done in a nonlinear model in an exact manner." (Pp. 120).

$$c_{A0} = \frac{F(\bar{X}_A\beta_A) - F(\bar{X}_B\beta_A)}{\sum_{k=1}^{K} \beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k})}.$$

When combined with the expression above, this is the estimated marginal effect for a change in $F(\tilde{X}\beta_A)$, where $\tilde{X}^T = (\tilde{X}_1, \ldots, \tilde{X}_k)$ is a vector in the interval between $\bar{X}_{A_k}$ and $\bar{X}_{B_k}$. Letting $\nabla F\tilde{X}\beta_A$ denote the vector of partial derivatives evaluated at the vector $\tilde{X}$, and noting that β_A is fixed, the detailed contributions to E are,

$$E_k = \nabla F(\tilde{X}\beta_A)^T(\bar{X}_{A_k} - \bar{X}_{B_k}) = \frac{F(\bar{X}_A\beta_A) - F(\bar{X}_B\beta_A)}{\sum_{k=1}^{K} \beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k})}\beta_{A_k}(\bar{X}_{A_k} - \bar{X}_{B_k})$$

It is easy to see that Bazen et al.'s result for the detailed composition components can be expressed in terms of Yun's composition weights, $W_{\Delta_{X_k}}$, multiplied by the difference in the actual and counterfactual prediction functions evaluated at covariate means.

$$E_k^{\text{MV}} = W_{\Delta_{X_k}}[F(\bar{X}_A\beta_A) - F(\bar{X}_B\beta_A)].$$

In contrast, Yun uses differences in the average actual and counterfactual predictions,

$$E_k^{\text{Yun}} = W_{\Delta_{X_k}}[\overline{F(X_A\beta_A)} - \overline{F(X_B\beta_A)}].$$

Unlike, Bazen, et al., Yun's approach is guaranteed to give components that sum to the observed difference in rates (as are Schwiebert's and Kaiser's).[24] Kaiser (2015) also invokes the mean value theorem in his derivation of the detailed composition effects, but his approach allows for group specific differences in the distributions of covariates; not only for differences in means, but for differences in higher moments (i.e., groupwise variability in the X_k).[25]

The C component is similarly motivated using the mean value theorem. Consider a vector $\tilde{\beta}$ between β_A and β_B. Applying MVT, and noting that $\bar{X}_B$ is fixed,

$$C_k = \nabla F(\bar{X}_B\tilde{\beta})^T(\beta_{A_k} - \beta_{B_k}) = \frac{F(\bar{X}_B\beta_A) - F(\bar{X}_B\beta_B)}{\sum_{k=1}^{K} \bar{X}_{B_k}(\beta_{A_k} - \beta_{B_k})}\bar{X}_{B_k}(\beta_{A_k} - \beta_{B_k}).$$

Variability in Decomposition Estimates

The characteristics and effects components provide no information about the precision of the contributions to group differences *per se*. For this reason, it is important to gauge the sampling variability (asymptotic variance) of E and C, as well as the detailed components, in substantive applications.

Because the components used in the decomposition are functions of maximum likelihood estimates, the delta method described by Rao (1973, Pp. 321-323)[26] can be used to derive asymptotic standard errors of the detailed contributions. Interval estimation and significance testing can be done in the usual way (see, e.g., Yun 2005).[27]

[24] This is not the case in several generalized linear models. For example, $\overline{F(X_j\beta_j}$ does not exactly equate to the mean response in probit and complementary log-log models fit to binomial/binary data.

[25] This approach is implemented in Kaiser's `Stata` module `glmdeco`.

[26] Rao, CR. 1973. *Linear Statistical Inference and Its Applications*. New York: Wiley.

[27] Yun, M-S. 2005a. "Hypothesis Tests when Decomposing Differences in the First Moment," *Journal of Economic and Social Measurement*, 30: 305-319

This approach utilizes expressions for the gradients of the detailed components with respect to the estimates, in addition to the variance/covariance matrix of the estimates from each group's hazard model, as we show next.

The endowment component is obtained as a weighted sum of the individual contributions, E_k,

$$E = \sum_{k=1}^{K} E_k = \sum_{k=1}^{K} W_{\Delta X_k} \{\overline{F(X_A\beta_A)} - \overline{F(X_B\beta_A)}. \tag{8}$$

Interval estimation and statistical hypothesis testing of the components of the detailed decomposition require computing the asymptotic variances of the E_k and C_k components appearing in the decomposition equation.

First we compute the gradient for E_k, i.e., $\partial E_k/\partial \beta_A^T$ which is a $1 \times K$ vector, the lth element of which is

$$\frac{\partial E_k}{\partial \beta_{A_l}} = W_{\Delta X_k} \left\{ \overline{\frac{\partial F(X_A\beta_A)}{\partial \beta_{A_l}}} - \overline{\frac{\partial F(X_B\beta_A)}{\partial \beta_{A_l}}} \right\} + \frac{\partial W_{\Delta_{X_k}}}{\partial \beta_{A_l}} \left\{ \overline{F(X_A\beta_A)} - \overline{F(X_B\beta_A)} \right\} \tag{9}$$

For the models considered in the `Stata` modules `mvdcmp` and `hazdcmp` and in the companion `GitHub` routines,

$$\frac{\partial F(X_j\beta_j)}{\partial \beta_l} = f(X_j\beta_j)X_{j_l} \qquad j = \{A, B\}.$$

Moreover, under Poisson sampling invoked by the piecewise constant exponential model, $F(.) = f(.)$.[28]

Letting $\mathbf{E} = (E_1, \ldots, E_K)$ denote the $K \times 1$ detailed characteristics effect vector and $\mathbf{\Sigma}_{\beta_A}$ denote the variance/covariance matrix of β_A, the asymptotic covariance matrix of the detailed characteristics component is

$$\mathbf{\Sigma_E} = \mathbf{G}_E \mathbf{\Sigma}_{\beta_A} \mathbf{G}_E^T,$$

where

$$\mathbf{G}_E = \left(\frac{\partial E_1}{\partial \beta_A^T}, \frac{\partial E_2}{\partial \beta_A^T}, \ldots, \frac{\partial E_K}{\partial \beta_A^T} \right)$$

is the $K \times K$ gradient matrix.

Following the same logic, the coefficient component can be written as the sum of individual contributions as,

$$C = \sum_{k=1}^{K} C_k = \sum_{k=1}^{K} W_{\Delta_{\beta_k}} \{\overline{F(X_B\beta_A)} - \overline{F(X_B\beta_B)}\}. \tag{10}$$

Each covariate's contribution to the overall coefficient component depends on the parameter vectors, β_A and β_B. The lth elements of the gradient for C_k are

$$\frac{\partial C_k}{\partial \beta_{A_l}} = W_{\Delta_{\beta_k}} \overline{f(X_B\beta_A)X_{B_l}} + \frac{\partial W_{\Delta_{\beta_k}}}{\partial \beta_{Al}} \left\{ \overline{F(X_B\beta_A)} - \overline{F(X_B\beta_B)} \right\}$$

and

$$\frac{\partial C_k}{\partial \beta_{B_l}} = \frac{\partial W_{\Delta_{\beta_k}}}{\partial \beta_{Bl}} \left\{ \overline{F(X_B\beta_A)} - \overline{F(X_B\beta_B)} \right\} - W_{\Delta_{\beta_k}} \overline{f(X_B\beta_B)X_{B_l}},$$

[28] It should be noted that the hazard-rate decomposition routines are not limited to continuous-time models, but also work for binary logit and complementary-log-log models suitable for discrete-time event history data using the functions `decomp.dtlogit` and `decomp.dtcloglog` (with the same arguments as `decomp.pwcexp`). In this case, the interval-width on an episode is assumed to be 1 (i.e., an offset = 0). But this can be changed by using a non-zero offset term to account for log exposure time in an interval. The results are on the probability scale rather than the rate scale unless actual exposure times are used.

Let $\mathbf{\Sigma}_{\beta_A}$ and $\mathbf{\Sigma}_{\beta_B}$ denote the covariance matrix of the estimates from the group A and B hazard regressions, and let $\mathbf{C} = C_1, \ldots, C_K$ denote the $K \times 1$ detailed coefficient effects vector. The asymptotic covariance matrix of the detailed coefficient components is then

$$\mathbf{\Sigma_C} = \mathbf{G}_{C_A}\mathbf{\Sigma}_{\beta_A}\mathbf{G}_{C_A}^T + \mathbf{G}_{C_B}\mathbf{\Sigma}_{\beta_B}\mathbf{G}_{C_B}^T$$

where

$$\mathbf{G}_{C_j} = \left(\frac{\partial C_1}{\partial \beta_j^T}, \frac{\partial C_2}{\partial \beta_j^T}, \ldots, \frac{\partial C_K}{\partial \beta_j^T}\right), \qquad j = \{A, B\}$$

is the $K \times K$ gradient matrix and $\dfrac{\partial C_k}{\partial \beta_j^T}$ is the $1 \times K$ gradient vector defined above.[29]

Appendix B: `Stata`

We briefly illustrate rate decomposition in `Stata`. However, because we rely on local macros and `mata`, it is not feasible to break up `code chunks` in the `Rmd` source file that produces this `pdf`.[30] Instead we run the whole batch at once and show the output only. To summarize, we read in the data, fit two rate models, show the output of a decomposition with Blacks as the reference, and fit two decompositions, extracting quantities that will be used to average the results in a `mata` routine, and post the average results back to `Stata`. We skip the descriptive results.

Hazard Models for Each Group

```
use hazdata24, clear
// rate models
glm devnt age11_16 age16_18 age18_20 age20_22 age22_24 age24_36 ///
   pctsmom nfamtran medu inc1000 nosibs magebir if blk==0, nocons f(p) offset(logexp) eform
est store hW
glm devnt age11_16 age16_18 age18_20 age20_22 age22_24 age24_36 ///
   pctsmom nfamtran medu inc1000 nosibs magebir if blk==1, nocons f(p) offset(logexp) eform
est store hB
esttab hW hB  using "tab1a.tex", ///
cells("b(fmt(3)) ci_l(fmt(2)) ci_u(fmt(2))") stats(N,fmt(0)) eform
// plot baseline hazards and hazard ratios
coefplot hW,  eform || hB, eform bylabels("White" "Black") xline(1)
// or
coefplot (hW, label("White"))  (hB, label("Black")),  eform   xline(1)
```

[29] Analytic derivatives are used in all routines.

[30] However all the `R` results can be reproduced using one dataset along with relevant `R` functions and the `Rmd` source.

Table 6: Baseline Hazards and Hazard Ratios of First Premarital Birth by Race

	Estimated Hazards and Hazard Ratios					
	White			Black		
	e^b	2.5%	97.5%	e^b	2.5%	97.5%
age11_16	0.006	0.00	0.01	0.014	0.01	0.02
age16_18	0.175	0.08	0.39	0.368	0.23	0.58
age18_20	0.244	0.11	0.55	0.455	0.29	0.73
age20_22	0.276	0.12	0.63	0.473	0.29	0.76
age22_24	0.165	0.07	0.40	0.348	0.21	0.58
age24_36	0.245	0.11	0.55	0.194	0.12	0.32
pctsmom	2.167	1.15	4.07	1.117	0.90	1.38
nfamtran	1.336	1.21	1.47	1.162	1.08	1.26
medu	0.873	0.83	0.92	0.937	0.91	0.96
inc1000	0.658	0.51	0.86	0.570	0.46	0.71
nosibs	1.182	1.10	1.27	1.053	1.02	1.09
magebir	0.949	0.93	0.97	0.971	0.96	0.98
N	10,131			5,325		

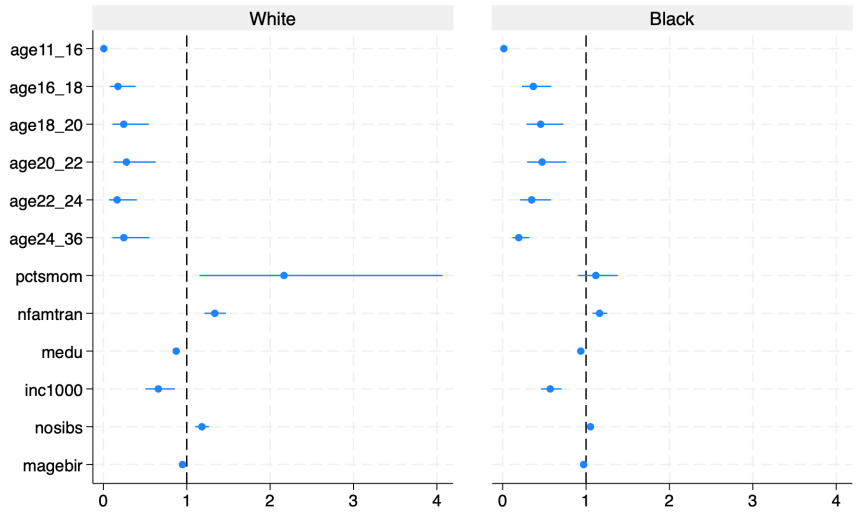


Figure 7: Coefficient Plots of Baseline Hazards and Hazard Ratios by Race

Decompose difference in Rates: Blacks as Group A

```
hazdcmp devnt  age11_16 age16_18 age18_20 age20_22 age22_24 age24_36 ///
pctsmom nfamtran medu inc1000 nosibs magebir, ///
     by(blk) id(iid) offset(logexp) scale(1000)
est store m1
```

Overall Decomposition

```
    Decomposition Results                    Split Episode Observations = 15,456
------------------------------------------------------------------------------------------------
```

```
        Reference group  (A):blk == 1             Rate = 25.30
        Comparison group (B):blk == 0             Rate = 4.82
------------------------------------------------------------------------------------------------
                        devnt|      Coef.  Std. Err.       z    P>|z|    [95% Conf. Interval]    Pct.
-----------------------------+------------------------------------------------------------------
                           E|    6.3504     0.8307    7.64    0.000      4.7222      7.9786    31.0
                           C|   14.1368     1.0958   12.90    0.000     11.9890     16.2847    69.0
-----------------------------+------------------------------------------------------------------
                        Total   20.4872
```

Detailed Decomposition

Due to Difference in Composition (E)

devnt	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]	Pct.
age11_16	-7.2992	0.9679	-7.54	0.000	-9.1963	-5.4020	-35.6
age16_18	-0.5649	0.1541	-3.66	0.000	-0.8671	-0.2628	-2.8
age18_20	0.4825	0.1610	3.00	0.003	0.1671	0.7980	2.4
age20_22	0.4709	0.1681	2.80	0.005	0.1414	0.8004	2.3
age22_24	0.6398	0.1785	3.58	0.000	0.2899	0.9897	3.1
age24_36	0.7060	0.1402	5.04	0.000	0.4313	0.9807	3.4
pctsmom	0.5576	0.5442	1.02	0.306	-0.5090	1.6243	2.7
nfamtran	0.5708	0.1553	3.68	0.000	0.2664	0.8752	2.8
medu	2.0380	0.5170	3.94	0.000	1.0247	3.0514	9.9
inc1000	7.0935	0.8700	8.15	0.000	5.3882	8.7987	34.6
nosibs	1.1915	0.4099	2.91	0.004	0.3882	1.9949	5.8
magebir	0.4637	0.1138	4.08	0.000	0.2407	0.6867	2.3

Due to Difference in Coefficients (C)

devnt	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]	Pct.
age11_16	2.0653	1.1995	1.72	0.085	-0.2856	4.4163	10.1
age16_18	1.5941	1.0230	1.56	0.119	-0.4109	3.5992	7.8
age18_20	1.0755	0.8305	1.29	0.195	-0.5523	2.7033	5.2
age20_22	0.6201	0.5668	1.09	0.274	-0.4909	1.7311	3.0
age22_24	0.5900	0.4195	1.41	0.160	-0.2322	1.4122	2.9
age24_36	-0.1290	0.2650	-0.49	0.626	-0.6485	0.3904	-0.6
pctsmom	-0.3456	0.1777	-1.95	0.052	-0.6938	0.0026	-1.7
nfamtran	-0.5923	0.2649	-2.24	0.025	-1.1115	-0.0730	-2.9
medu	7.2695	2.8863	2.52	0.012	1.6124	12.9267	35.5
inc1000	-1.2491	1.5085	-0.83	0.408	-4.2058	1.7075	-6.1
nosibs	-1.9146	0.6676	-2.87	0.004	-3.2231	-0.6060	-9.3
magebir	5.1529	3.1917	1.61	0.106	-1.1028	11.4086	25.2

Coefficient Plots

```
// coefficient plot by component (scaled)
coefplot m1, keep(E:)  transform(*=@*1000) || ///
         m1, keep(C:)  transform(*=@*1000) ///
         xline(0) bylabels("E-Component" "C-Component") ///
         byopts(xrescale)
```

Set-up Average Decomposition

```
// helper program for combined decompositions
cap program drop add_cmd
program add_cmd, eclass
     args cmd
   eret local cmd "`cmd'"
end

// decompose difference in rates from Black perspective: Blacks are group (A)
hazdcmp devnt  age11_16 age16_18 age18_20 age20_22 age22_24 age24_36 ///
pctsmom nfamtran medu inc1000 nosibs magebir, ///
     by(blk) id(iid) offset(logexp) scale(1000)
```

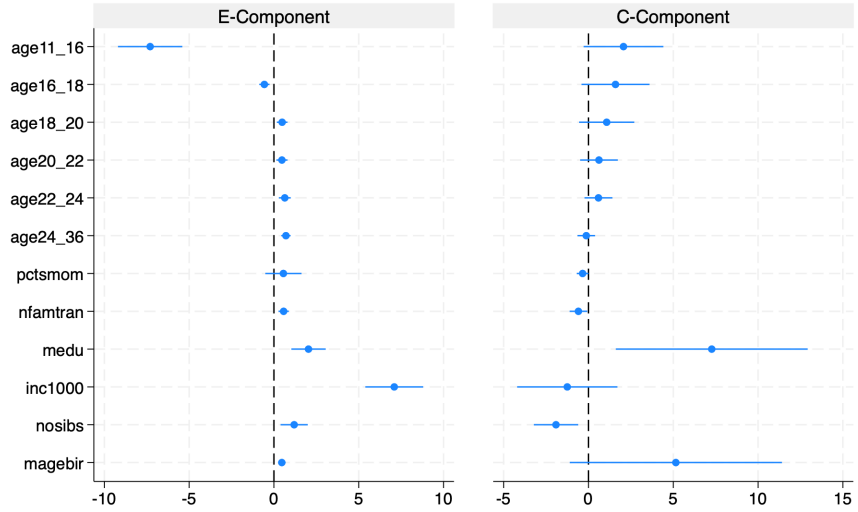


Figure 8: Coefficient Plots: Detailed Decomposition

```
local vnames  "`e(cvarlist)'"   // get list of varnames appearing in the output
// extract returns from model
mat bA = e(b)
mat VA = e(V)

// decompose difference in rates from White perspective: Whites are group (A)
hazdcmp devnt age11_16 age16_18 age18_20 age20_22 age22_24 age24_36 ///
 pctsmom nfamtran medu inc1000 nosibs magebir, ///
      by(blk) id(iid) offset(logexp) scale(1000) reverse
// extract returns from model
mat bB = e(b)
mat VB = e(V)
```

Mata Routine for Combining Results from Two Decompositions

```
mata:
 b0   = st_matrix("bA")
 V0   = st_matrix("VA")
 b1   = st_matrix("bB")
 V1   = st_matrix("VB")
  s   = 1000  // scale factor
// sign change for averaging
  b    =  s * (b0 - b1)/2
  V    = s^2 * (V0 + V1):/4
  k    = cols(b)
  bE   = sum(b[1..k/2])
  bC   = sum(b[k/2..k])
  VE   = sum(V[1..k/2,1..k/2])
  VC   = sum(V[k/2..k, k/2..k])
  bR   = bE, bC, (bE + bC)
  T    = VE, VC, (VE + VC)
  VR   = diag(T)
st_matrix("b", b)
```

```
st_matrix("V", V)
st_matrix("bR", bR)
st_matrix("VR", VR)
st_matrix("VR", VR)
end
```

Post and Store Results

```
local smrynames = "E C Total"
// post results
matrix colnames b = `vnames'
matrix colnames V = `vnames'
matrix rownames V = `vnames'
matrix colnames bR = `smrynames'
matrix colnames VR = `smrynames'
matrix rownames VR = `smrynames'
// associate the -mata- returns with a command name
// Overall
eret post bR VR, depname("Total")
add_cmd "hazdcmp"
est store mTot
// standard output
eret display,  cformat(%6.5f) level(95)
// estout with added column for pct
mata:
b    = st_matrix("e(b)")
sumb = sum(b)
pct  = b:/b[3] * 100
st_matrix("pct", pct)
end
mat colnames pct = `smrynames'
estadd matrix pct = pct
esttab  mTot using "tab3a.tex", replace ///
cells("b(fmt(3)) ci_l(fmt(2)) ci_u(fmt(2)) pct(fmt(1))")
```

```
// Detailed
eret post b V , depname("Averaged")
add_cmd "hazdcmp"
est store mAve
// standard output
eret display, cformat(%6.5f) level(95)
// estout output with added column for pct
mata:
b    = st_matrix("e(b)")
sumb = sum(b)
pct  = b:/sumb * 100
st_matrix("pct", pct)
end
mat colnames pct = `vnames'
estadd matrix pct = pct
estout mAve using "Tab2a.tex", replace ///
cells("b(fmt(3)) ci_l(fmt(2)) ci_u(fmt(2)) pct(fmt(1))") unstack
```

Table 7: Decompostion of Racial Difference in 1st Premarital Birth Rates: Averaged

	Components of Difference in Rates							
	E-Component	2.5%	97.5%	pct	C-Component	2.5%	97.5%	pct
age11_16	-5.021	-6.05	-3.99	-24.5	2.752	0.49	5.02	13.4
age16_18	-0.436	-0.61	-0.27	-2.1	1.968	0.18	3.75	9.6
age18_20	0.376	0.20	0.55	1.8	1.188	-0.09	2.47	5.8
age20_22	0.362	0.18	0.55	1.8	0.660	-0.18	1.50	3.2
age22_24	0.490	0.29	0.69	2.4	0.601	0.01	1.19	2.9
age24_36	0.447	0.30	0.60	2.2	-0.131	-0.50	0.24	-0.6
pctsmom	0.886	0.10	1.67	4.3	-1.110	-2.06	-0.17	-5.4
nfamtran	0.457	0.29	0.62	2.2	-0.810	-1.33	-0.29	-4.0
medu	1.678	1.10	2.25	8.2	8.112	3.63	12.59	39.6
inc1000	4.369	3.38	5.36	21.3	-1.091	-2.94	0.75	-5.3
nosibs	1.195	0.68	1.71	5.8	-2.816	-4.21	-1.42	-13.7
magebir	0.362	0.23	0.49	1.8	5.999	0.81	11.18	29.3
Total	5.165	3.94	6.39	24.8	15.684	13.94	17.43	75.2

Coefficient Plot

```
// use newly created model objects for coef plots

coefplot mAve, keep(E:)  || ///
         mAve, keep(C:)     ///
         xline(0) bylabels("E-Component" "C-Component") ///
         byopts(xrescale)
```

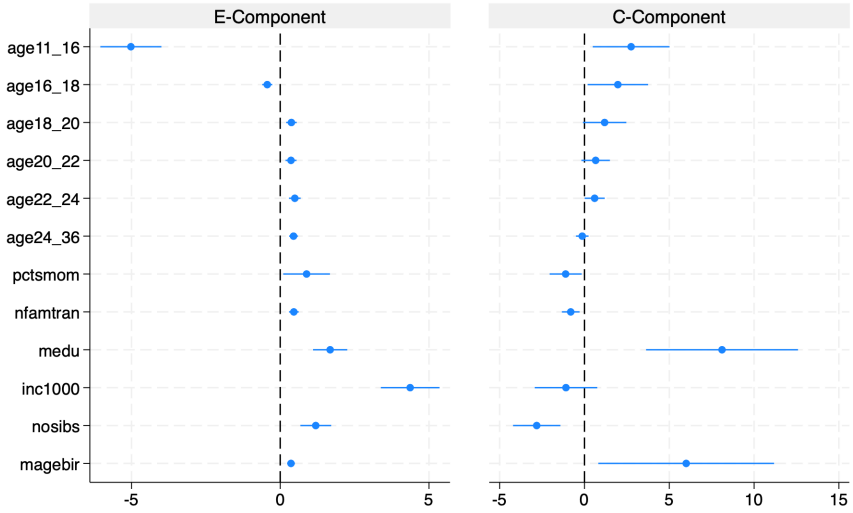


Figure 9: Stata: Hazard Rate Decomposition Results (Averaged)

Appendix C: Baseline Hazard

How should we interpret the detailed contributions of the baseline hazard? Recall that the hazard models used for each group are defined on exogenously chosen time intervals of risk that are common across the groups whose overall rate differential is being compared. The split-episode data structure generates n_i contributions ($\Delta t_{i\ell},\ \ell = 1, \ldots, n_i$) for individual i's risk, where n_i ranges from the first interval to the last interval that an individual was at risk of the event of interest. Individuals are removed from risk either by the occurrence of the event of interest, by censoring due to another event (i.e., marriage, in the case of 1st premarital birth), or by having not experienced the event of interest by the end of the observation period (i.e., the fertility histories span a period from 1979 to 1992. The youngest respondent in 1979 is 14 years old and the oldest respondent in 1992 is 35 years old. We don't observe fertility histories beyond 1992.)

Contribution of the Baseline Hazard to the Composition Component, E: A standard multivariate decomposition requires that regression models for each group contain an intercept. Because the mean of the constant vector is equal to 1 across groups, the difference in means is zero and the associated contribution of the intercept to the composition component (E) is 0. However, this is not the case for the piecewise constant hazard models considered here. The standard intercept has been replaced by $\max(n_i), i = 1, \ldots, N$ intercepts that correspond to the number of user-defined episodes in the split-episode data. The contributions to E for the pieces of the baseline hazard should be considered as placeholders (i.e., for accounting purposes) to ensure that the detailed characteristics contributions (E_k) sum to the overall contribution E. That is, they play the role of the usual constant that is necessary for the model predictions to sum to the observed rate. The individual contributions in some sense reflect group differences in the exposure to risk in the time intervals under study, with the realization that these differences reflect unique event and censoring patterns specific to each group.

Contribution of the Baseline Hazard to the Composition Component, C: Because C reflects the contribution of differences in β, his component has more relevance as it reflects a contribution of group differences in interval-specific log-hazards. Note that this component depends on the sample proportions in each time interval (i.e., the means of the dummy variables D_ℓ.)

In the detailed decomposition presented in Table 5, the sum of the contributions of the baseline hazard to C is 7.74, ranging from 2.74 in the 1st age interval to 0.601 in the last age interval. These contributions reflect the Black-White differences in age-specific log hazards of 1st premarital birth. For example, shifting the Black baseline hazard in the age 11-16 interval that of Whites (i.e., cutting it more than half from 0.014 to 0.006, as shown in Table 3 would be expected to reduce the premarital birth rate gap by roughly 3 births per 1,000 (or 13%). Doing the same for the age 16-18 (i.e., from .386 to .175) would be expected to reduce the premarital birth rate gap by about 2 births per 1,000 (or 10%). The remaining terms are statistically 0.

Appendix D: An Identification Problem

A well-known identification problem exists when using factor variables.[31] This affects only the detailed decomposition for the unexplained component (the C_k's). Specifically, the total contribution of a factor variable differs depending on the choice of the reference category. A solution to this problem is to use effect coding for factor variables. This forces a sum-to-zero constraint on the coefficients over all factor levels. Suppose we coded maternal education into three categories, < 12, 12, and > 12 as follows:

```
#
# construct a factor for R mother's educational level
#
df <-  mutate(df, medcat = ifelse(medu < 12, 1,
                             ifelse(medu == 12, 2,
                               ifelse(medu > 12, 3, NA))),
```

[31] Yun, M-S. 2005b. "A Simple Solution to the Identification Problem in Detailed Wage Decompositions," *Economic Inquiry*, 43: 766-772, with Erratum, *Economic Inquiry*, 44: 198. Yun, M-S. 2008. "Identification Problem and Detailed Oaxaca Decomposition: A General Solution and Inference," *Journal of Economic and Social Measurement*, 33: 27-38.

```
                medcat = factor(medcat,
                                labels=c("<12", "12", ">12")))
```

Use the `contrastable` library to convert `medcat` from dummy-variable to centered-effects coding (i.e., sum-to-zero constraint) in the original person-level data.

```
#
library(contrastable)
df <- set_contrasts(df, medcat ~ sum_code)
```

The new codings are,

```
$medcat
    12 >12
<12 -1 -1
12   1  0
>12  0  1
```

Pass the data frame on to episode splitting and on to the survey design setup. With < 12 as the omitted category, the total maternal education contribution to the overall coefficients component is $C_{\text{medcat12}} + C_{\text{medcat}>12}$, and is invariant with respect to choice of omitted category. The example below shows this using a simpler model.

< 12 Omitted Category in Dummy-Variable Coding

```
     medcat12  0.7041091  0.2595921     2.7124       +3.4
    medcat>12 -0.0318791  0.0810364    -0.3934       -0.2
```

> 12 Omitted Category in Dummy-Variable Coding

```
      medcat12  0.8636733  0.3184206     2.7124       +4.2
     medcat>12 -0.1914434  0.0569471    -3.3618       -0.9
```